

Test Exam on Foundations of Neuro-Symbolic AI
July 6, 2026
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Name: _____

Matrikel number: _____

Grading policy

Choose the right answers to the questions. There is at least one right answer to each question, and some questions have multiple right answers.

- For each right answer you will get one point.
- For each wrong answer you will loose one point.
- Each question will be graded with at least 0 points.

You are allowed to bring one handwritten A4 sheet (both pages).

1. Let us consider the propositional formula

$$f[X_0, X_1] = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} [X_0, X_1] .$$

How many coordinates does this tensor have?

- ☐ 4 ☐ 16 ☐ 2

What is the order of this tensor?

- ☐ 4 ☐ 16 ☐ 2

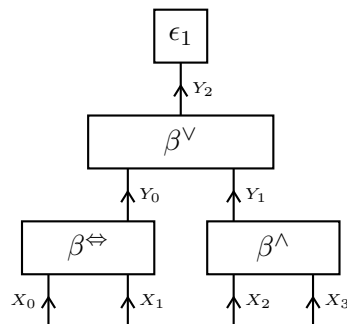
2. How many coordinates does a propositional formula of d variables have?

- ☐ d^2 ☐ 2^d ☐ d^d ☐ $2^{(2^d)}$

3. Which of these is a syntactical representation of the formula in question 1?

- ☐ $X_0 \Leftrightarrow X_1$ ☐ $X_0 \Leftrightarrow \neg X_1$ ☐ $X_0 \Rightarrow X_1$

4. Consider the tensor network diagram:



Which of these formulas is equivalent to the contraction of this tensor network leaving only X_0, X_1, X_2, X_3 open?

☐ $(X_0 \Leftrightarrow (X_1 \vee X_3)) \wedge X_3$ ☐ $Y_2 \Leftrightarrow (Y_0 \vee Y_1)$ ☐ $(X_0 \Leftrightarrow X_1) \vee (X_2 \wedge X_3)$

5. Which of these matrices encodes the constraint that the variables X_0 and X_1 of dimension 3 are not in the same state?

☐ $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} [X_0, X_1]$ ☐ $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} [X_0, X_1]$ ☐ $\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} [X_0, X_1]$

6. Let there be two formulas

$$f_0[X_0, X_1] = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} [X_0, X_1] \quad \text{and} \quad f_1[X_0, X_1] = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} [X_0, X_1]$$

Which of these entailment statements are true?

☐ $f_1 \models f_0$ ☐ $f_0 \models f_1$ ☐ $\neg f_1 \models \neg f_0$ ☐ $\neg f_0 \models \neg f_1$ ☐ none of the before

7. Which of these tensors is a probability distribution?

☐ $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} [X_0, X_1]$ ☐ $\begin{pmatrix} 0.2 & 0.2 & 0.3 \\ 0.1 & 0.5 & 0.8 \end{pmatrix} [X_0, X_1]$ ☐ $\begin{pmatrix} 0.1 & 0.1 & 0.3 \\ 0.2 & 0.2 & 0.1 \end{pmatrix} [X_0, X_1]$

8. Let $\mathbb{P}[X_{[d]}]$ be a distribution of the variables $X_{[d]}$ and $\mathcal{U} \subset [d]$. Which of these contractions computes the marginal distribution of the variables $X_{\mathcal{U}}$?

☐ $\langle \mathbb{P}[X_{[d]}] \rangle_{[\emptyset]}$ ☐ $\langle \mathbb{P}[X_{[d]}] \rangle_{[X_{[d]}/\mathcal{U}]}$ ☐ $\langle \mathbb{P}[X_{[d]}] \rangle_{[X_{\mathcal{U}}]}$

9. Consider a distribution

$$\mathbb{P}[X_0, X_1, X_2] = \begin{pmatrix} 0.20 & 0.05 \\ 0.15 & 0.10 \end{pmatrix} \begin{pmatrix} 0.20 & 0.05 \\ 0.15 & 0.10 \end{pmatrix} [X_0, X_1, X_2]$$

Which of these is the distribution $\mathbb{P}[X_0, X_1 | X_2]$ of X_0, X_1 conditioned on X_2 ?

☐ $\begin{pmatrix} 0.20 & 0.05 \\ 0.15 & 0.10 \end{pmatrix} [X_0, X_1]$

☐ $\begin{pmatrix} 0.40 & 0.10 \\ 0.30 & 0.20 \end{pmatrix} \begin{pmatrix} 0.40 & 0.10 \\ 0.30 & 0.20 \end{pmatrix} [X_0, X_1, X_2]$

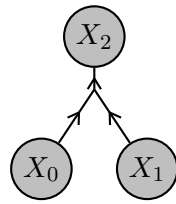
☐ $\begin{pmatrix} 0.20 & 0.05 \\ 0.15 & 0.10 \end{pmatrix} \begin{pmatrix} 0.20 & 0.05 \\ 0.15 & 0.10 \end{pmatrix} [X_0, X_1, X_2]$

Is $(X_0 \perp X_2)$ true with respect to this distribution?

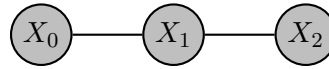
☐ Yes ☐ No

10. Consider these two hypergraph:

a)



b)



Which of these independence statement is true with respect to any Bayesian network on the hypergraph a)?

- ☐ $(X_0 \perp X_1)$ ☐ $(X_2 \perp X_0, X_1)$ ☐ $(X_2 \perp X_0) \mid X_1$

Which of these independence statement is true with respect to any Markov network on the hypergraph b)?

- ☐ $(X_0 \perp X_2)$ ☐ $(X_2 \perp X_0) \mid X_1$ ☐ $(X_0 \perp X_2) \mid X_1$